

1. Student Academic Performance Dashboard

Code :

```
import pandas as pd
import matplotlib.pyplot as plt

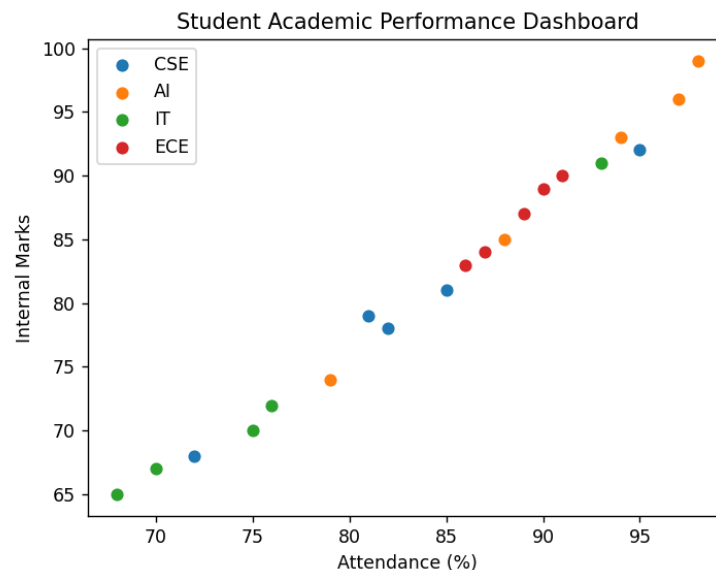
df =
pd.read_csv("C:/Users/naras/Downloads/DSA0602/student_academic_performance.csv")

colors = {"CSE":"blue","AI":"green","IT":"red","ECE":"orange"}

for dept in df["Department"].unique():
    d = df[df["Department"] == dept]
    plt.scatter(d["Attendance (%)"], d["Internal Marks"], label=dept)

plt.title("Student Academic Performance Dashboard")
plt.xlabel("Attendance (%)")
plt.ylabel("Internal Marks")
plt.legend()
plt.show()
```

Output :



2. Electric Vehicle Market Analysis

Code :

```
import pandas as pd
import matplotlib.pyplot as plt
```

```

df =
pd.read_csv("C:/Users/naras/Downloads/DSA0602/electric_vehicle_mar
ket_analysis.csv")

plt.figure(figsize=(10,6))

for manufacturer in df["Manufacturer"].unique():
    data = df[df["Manufacturer"] == manufacturer]

    plt.scatter(
        data["Battery Capacity (kWh)"],
        data["Price"],
        s=data["Driving Range"], # Bubble size
        alpha=0.6,
        label=manufacturer
    )

plt.title("Electric Vehicle Market Analysis")
plt.xlabel("Battery Capacity (kWh)")
plt.ylabel("Price")

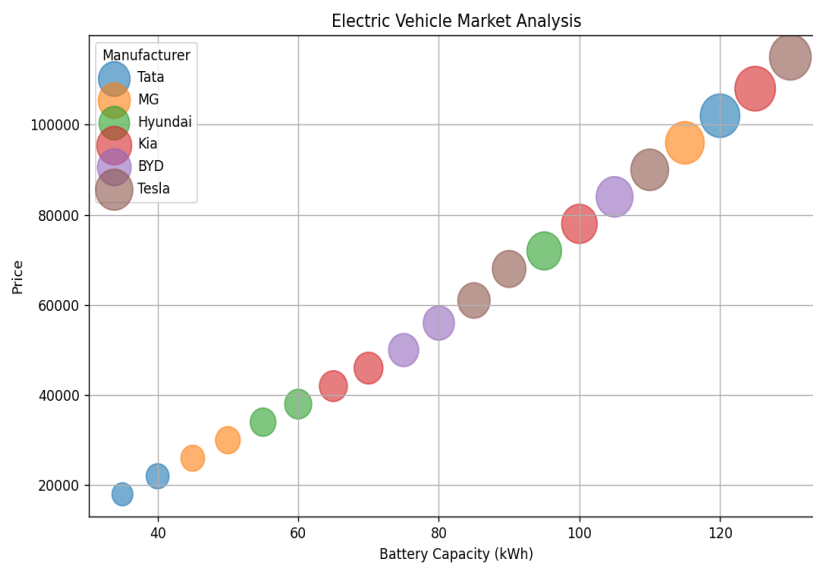
plt.legend(title="Manufacturer")

plt.grid(True)

plt.show()

```

Output:



3. Hospital Patient Admission Analysis

Code :

```
import pandas as pd
import matplotlib.pyplot as plt

df =
pd.read_csv("C:/Users/naras/Downloads/DSA0602/hospital_patient_admission_analysis.csv")

plt.figure(figsize=(10,6))

bars = plt.bar(
    df["Department"],
    df["Number of Patients"],
    color=["blue","green","red","orange","purple",
          "cyan","pink","brown","gray","gold"]
)

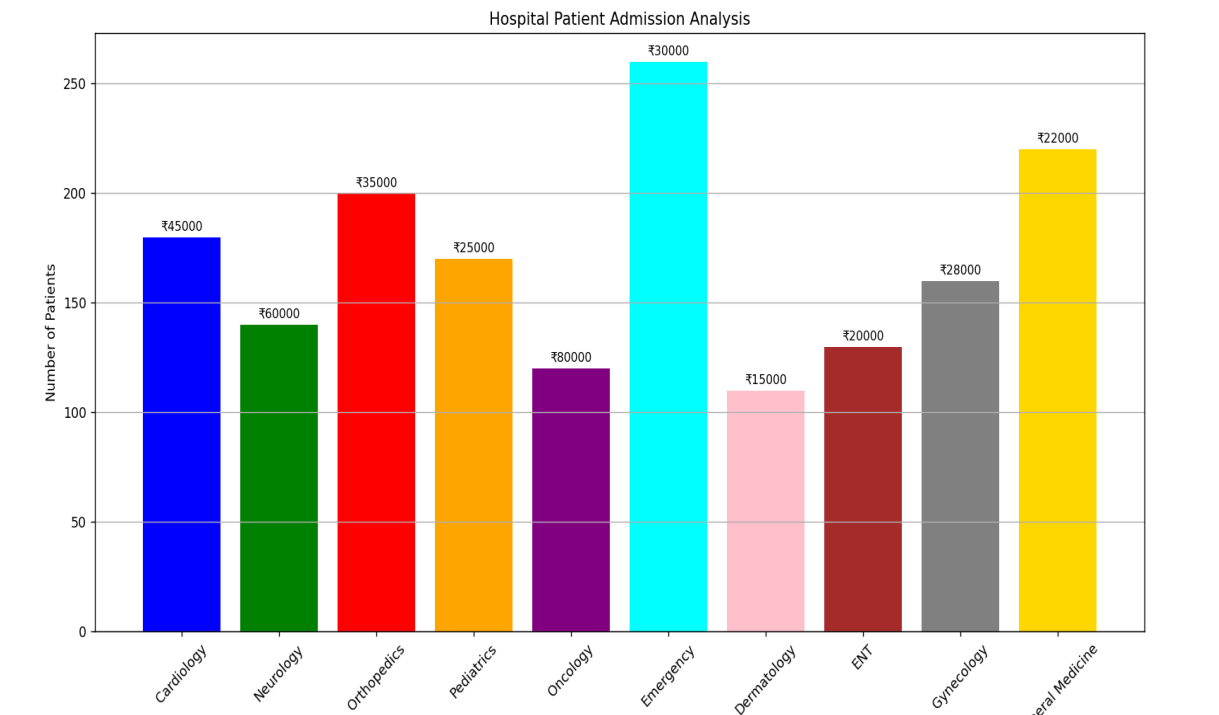
for bar, cost in zip(bars, df["Average Treatment Cost"]):
    plt.text(
        bar.get_x() + bar.get_width()/2,
        bar.get_height() + 3,
        f"₹{cost}",
        ha="center",
        fontsize=9
    )

plt.title("Hospital Patient Admission Analysis")
plt.xlabel("Department")
plt.ylabel("Number of Patients")

plt.xticks(rotation=45)
plt.grid(axis='y')

plt.show()
```

Output :



4. Weather Monitoring System

Code :

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("weather_monitoring_system.csv")

df["Date"] = pd.to_datetime(df["Date"])

plt.figure(figsize=(10,6))

for city, color, style in zip(
    df["City"].unique(),
    ["blue", "green", "red"],
    ["-", "--", ":"]
):
    city_data = df[df["City"] == city]
    plt.plot(
        city_data["Date"],
```

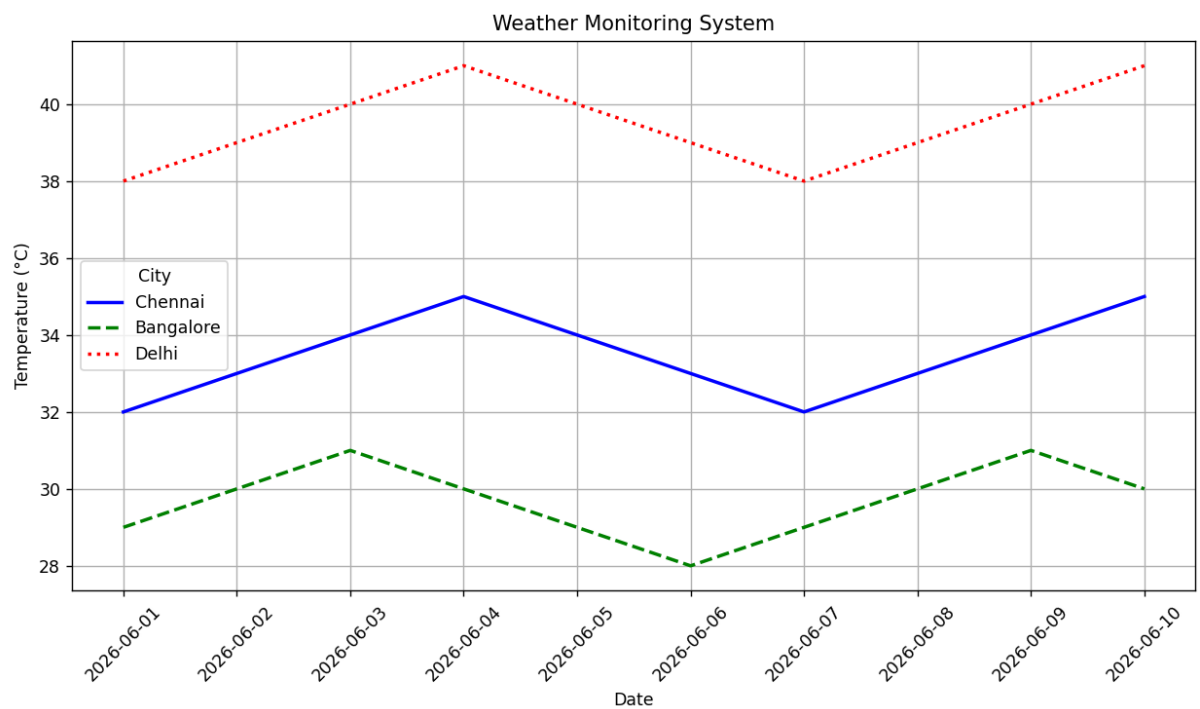
```

city_data["Temperature"],
color=color,
linestyle=style,
linewidth=2,
label=city
)

plt.title("Weather Monitoring System")
plt.xlabel("Date")
plt.ylabel("Temperature (°C)")
plt.legend(title="City")
plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()
plt.show()

```

Output :



5. Supermarket Sales Analysis

Code :

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

df =
pd.read_csv("C:/Users/naras/Downloads/DSA0602/supermarket_sales_an
alysis (1).csv")

df = df[df["Month"] == "January"]

categories = df["Product Category"].unique()
branches = df["Branch"].unique()

x = np.arange(len(categories))
width = 0.25

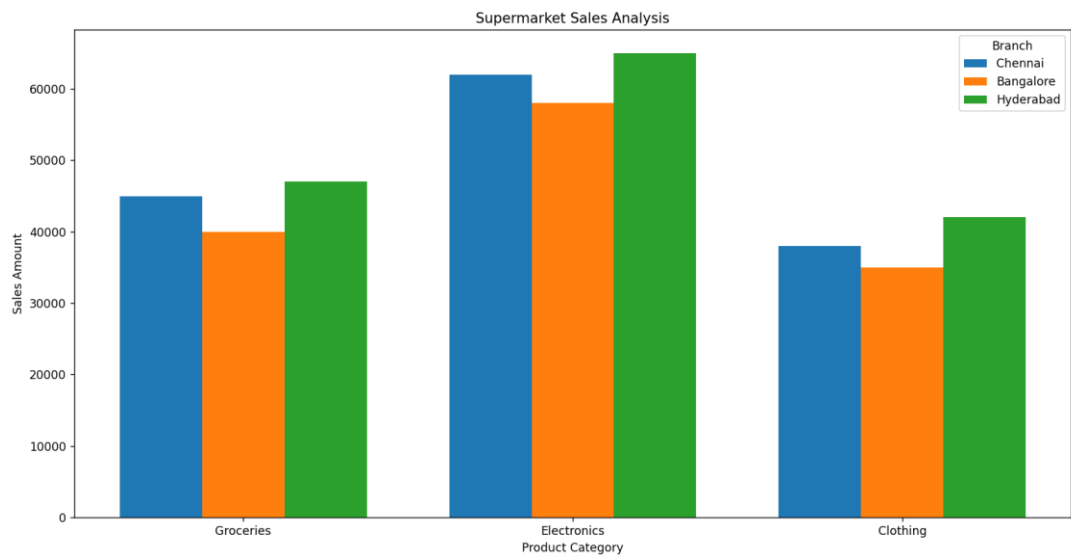
plt.figure(figsize=(8,6))

for i, branch in enumerate(branches):
    sales = df[df["Branch"] == branch]["Sales Amount"].values
    plt.bar(x + i*width, sales, width, label=branch)

plt.xticks(x + width, categories)
plt.xlabel("Product Category")
plt.ylabel("Sales Amount")
plt.title("Supermarket Sales Analysis")
plt.legend(title="Branch")

plt.show()
```

Output :



6. Employee Salary Distribution

Code :

```
import pandas as pd
import matplotlib.pyplot as plt

df =
pd.read_csv("C:/Users/naras/Downloads/DSA0602/employee_salary_distribution.csv")

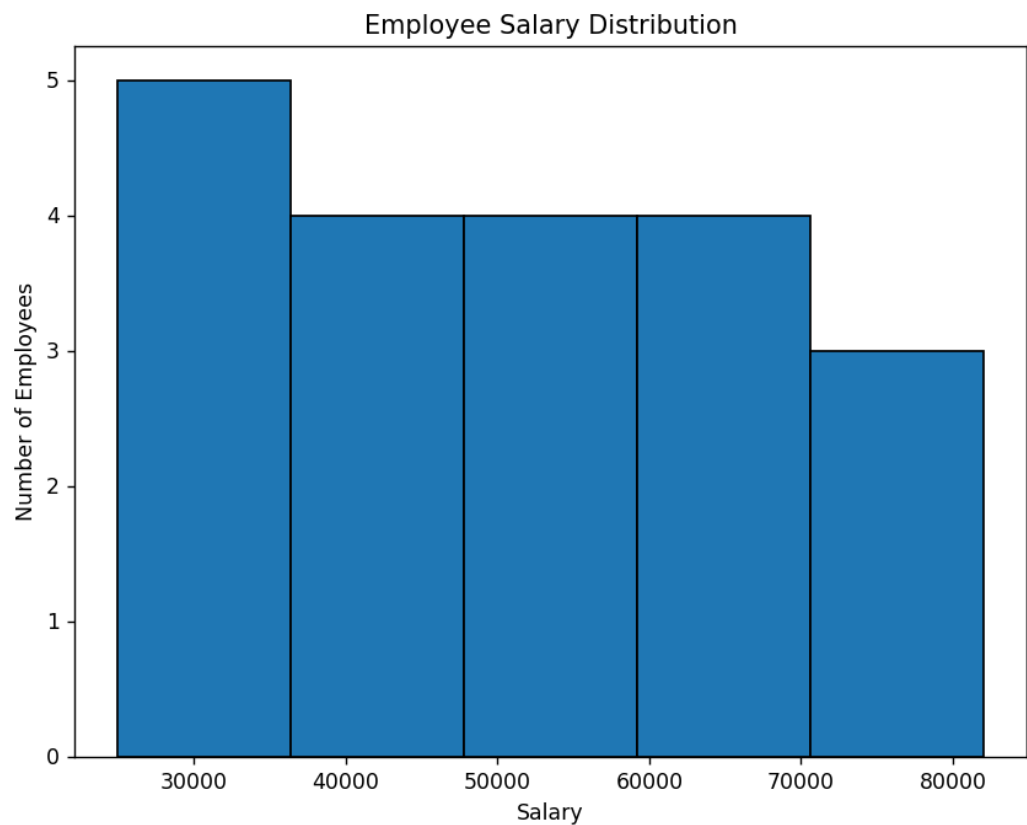
plt.figure(figsize=(8,6))

plt.hist(df["Salary"], bins=5, edgecolor="black")

plt.title("Employee Salary Distribution")
plt.xlabel("Salary")
plt.ylabel("Number of Employees")

plt.show()
```

Output :



7. Smartphone Market Comparison

Code :

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("smartphone_market.csv")

markers = {
    "Samsung": "o",
    "Apple": "s",
    "Xiaomi": "^",
    "OnePlus": "D",
    "Realme": "P",
    "Vivo": "X",
    "Oppo": "*",
    "Motorola": "v",
    "Google": "<",
    "Nothing": ">",
    "iQOO": "h",
```



```
"ASUS": "8",  
"Nokia": "p",  
"Sony": "d",  
"Huawei": "+"  
}
```

```
plt.figure(figsize=(12, 8))
```

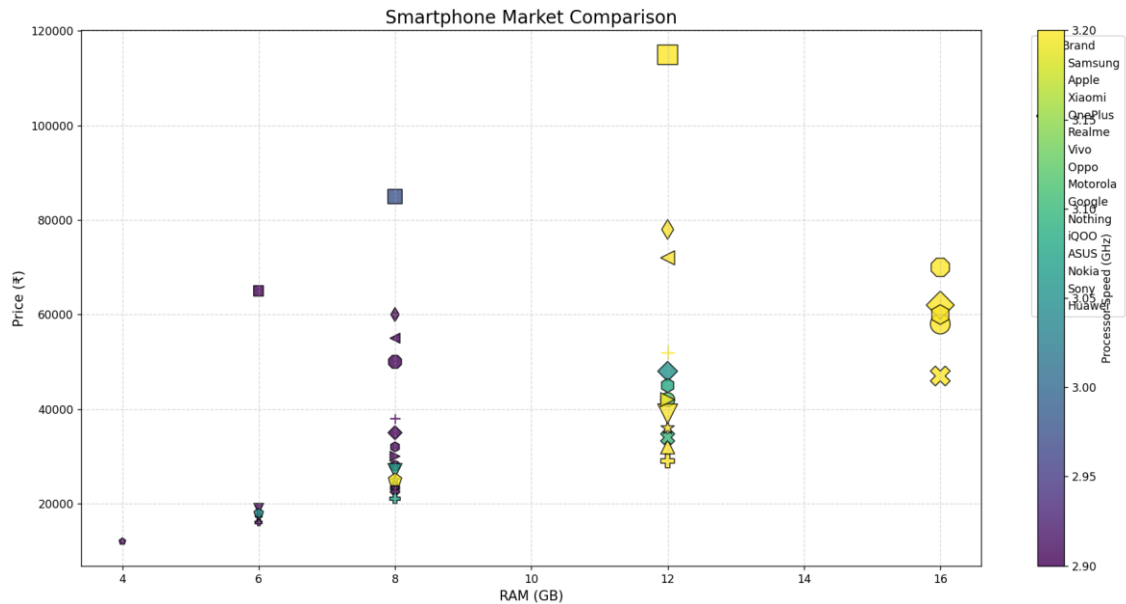
```
for brand in df["Brand"].unique():  
    brand_data = df[df["Brand"] == brand]
```

```
    scatter = plt.scatter(  
        brand_data["RAM"],  
        brand_data["Price"],  
        s=brand_data["Storage"] * 0.6,  
        c=brand_data["Processor_Speed"],  
        cmap="viridis",  
        marker=markers.get(brand, "o"),  
        edgecolors="black",  
        alpha=0.8,  
        label=brand  
    )
```

```
cbar = plt.colorbar(scatter)  
cbar.set_label("Processor Speed (GHz)")
```

```
plt.title("Smartphone Market Comparison")  
plt.xlabel("RAM (GB)")  
plt.ylabel("Price (₹)")  
plt.legend(title="Brand", bbox_to_anchor=(1.05, 1), loc="upper left")  
plt.grid(True)  
plt.tight_layout()  
plt.show()
```

Output :



8. Online Movie Rating Analysis

Code :

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("C:/Users/naras/Downloads/DSA0602/movie_ratings.csv")

heatmap_data = df.pivot(index="Genre", columns="Release_Year",
values="Average_Rating")

plt.figure(figsize=(8, 6))

plt.imshow(heatmap_data, cmap="YlOrRd", aspect="auto")

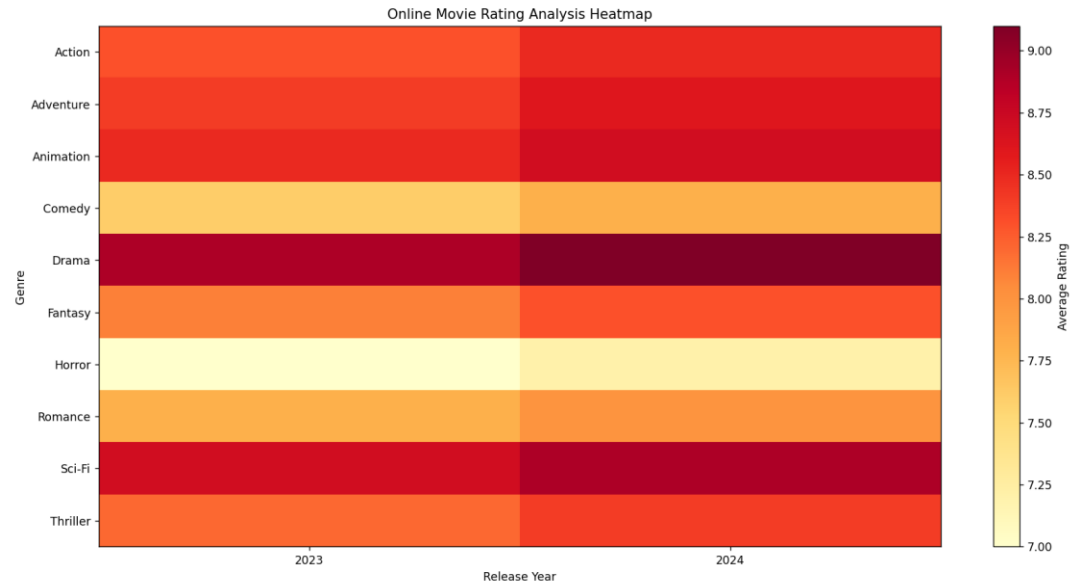
plt.colorbar(label="Average Rating")

plt.xticks(range(len(heatmap_data.columns)), heatmap_data.columns)
plt.yticks(range(len(heatmap_data.index)), heatmap_data.index)

plt.title("Online Movie Rating Analysis Heatmap")
plt.xlabel("Release Year")
plt.ylabel("Genre")
```

```
plt.tight_layout()
plt.show()
```

Output :



9. COVID-19 State-wise Analysis

Code :

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("C:/Users/naras/Downloads/DSA0602/covid_state_data.csv")

heatmap_data = df.set_index("State")[["Active_Cases"]]

plt.figure(figsize=(6, 10))

plt.imshow(heatmap_data, cmap="Reds", aspect="auto")

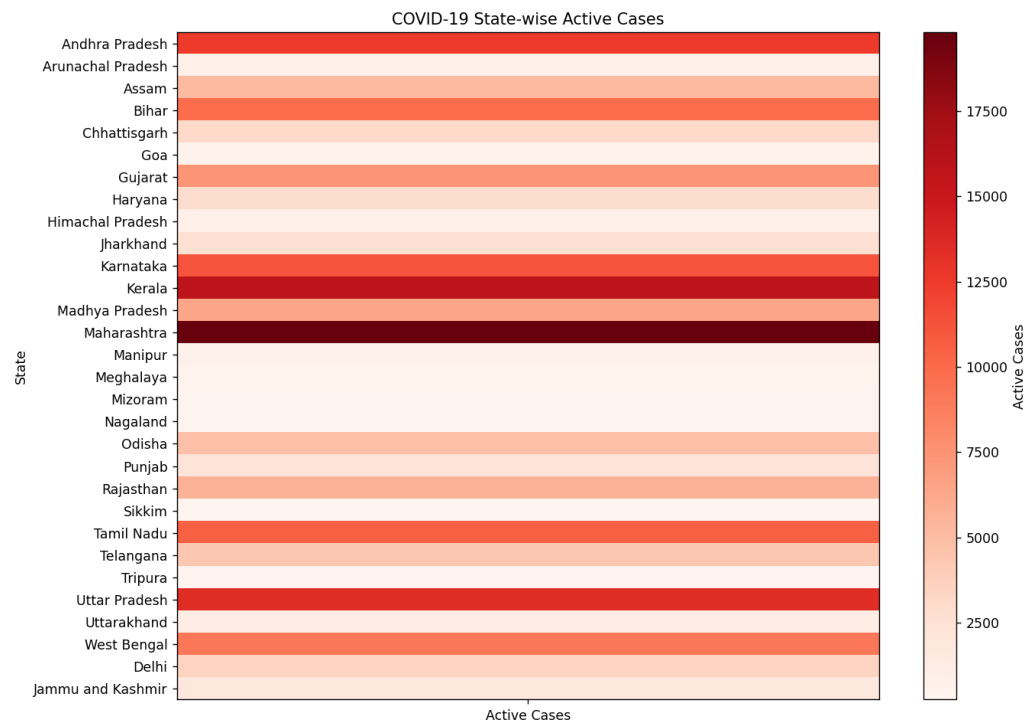
plt.colorbar(label="Active Cases")

plt.xticks([0], ["Active Cases"])
plt.yticks(range(len(heatmap_data.index)), heatmap_data.index)

plt.title("COVID-19 State-wise Active Cases")
plt.xlabel("")
plt.ylabel("State")
```

```
plt.tight_layout()
plt.show()
```

Output :



10. Iris Flower Classification

Code :

```
from sklearn.datasets import load_iris
import pandas as pd
import matplotlib.pyplot as plt

iris = load_iris()

df = pd.DataFrame(iris.data, columns=iris.feature_names)
df["Species"] = [iris.target_names[i] for i in iris.target]

markers = {
    "setosa": "o",
    "versicolor": "s",
    "virginica": "^"
}

plt.figure(figsize=(10, 7))
```

```

for species in df["Species"].unique():
    data = df[df["Species"] == species]

    scatter = plt.scatter(
        data["sepal length (cm)"],
        data["petal length (cm)"],
        s=data["sepal width (cm)] * 50,
        c=[iris.target_names.tolist().index(species)] * len(data),
        cmap="viridis",
        marker=markers[species],
        edgecolors="black",
        alpha=0.8,
        label=species
    )

plt.colorbar(scatter, ticks=[0, 1, 2], label="Species")
plt.clim(-0.5, 2.5)

plt.title("Iris Flower Classification")
plt.xlabel("Sepal Length (cm)")
plt.ylabel("Petal Length (cm)")
plt.legend(title="Species")
plt.grid(True)
plt.show()

```

Output :

